

Customer Support AI

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Overview

Customer Support AI is a modular retrieval-based customer support assistant. Instead of generating unrestricted answers, it searches an approved CSV knowledge base using semantic sentence embeddings and returns the closest verified response.

Theory

Every approved FAQ question is encoded into a dense embedding using a SentenceTransformer model. User questions are embedded with the same model and compared using cosine similarity. The highest-scoring approved question is selected. If the similarity score is below the configured threshold, the system safely falls back instead of inventing information.

Sentiment Analysis

A Hugging Face sentiment model classifies each message as positive, neutral, or negative. A configurable escalation threshold recommends transferring strongly negative conversations to a human support agent.

Training

Load a CSV knowledge base, choose embedding and sentiment models, configure thresholds, click Train / Build Embeddings, then optionally save the generated training artifact. Later sessions can load the saved artifact automatically.

GUI Guide

The left panel is used for training and configuration. Browse a CSV, select AI models, adjust thresholds, build embeddings, save or reload training artifacts. The Frontend tab is the customer-facing chat interface. The Backend tab exposes similarity score, matched FAQ, sentiment, escalation recommendation, processing times, statistics, conversation history, and the sentiment meter.

Architecture

The project is organized into modules responsible for CSV loading, embedding generation, semantic retrieval, sentiment analysis, conversation logging, persistent artifacts, and the graphical interface. This separation keeps the code maintainable and reusable.

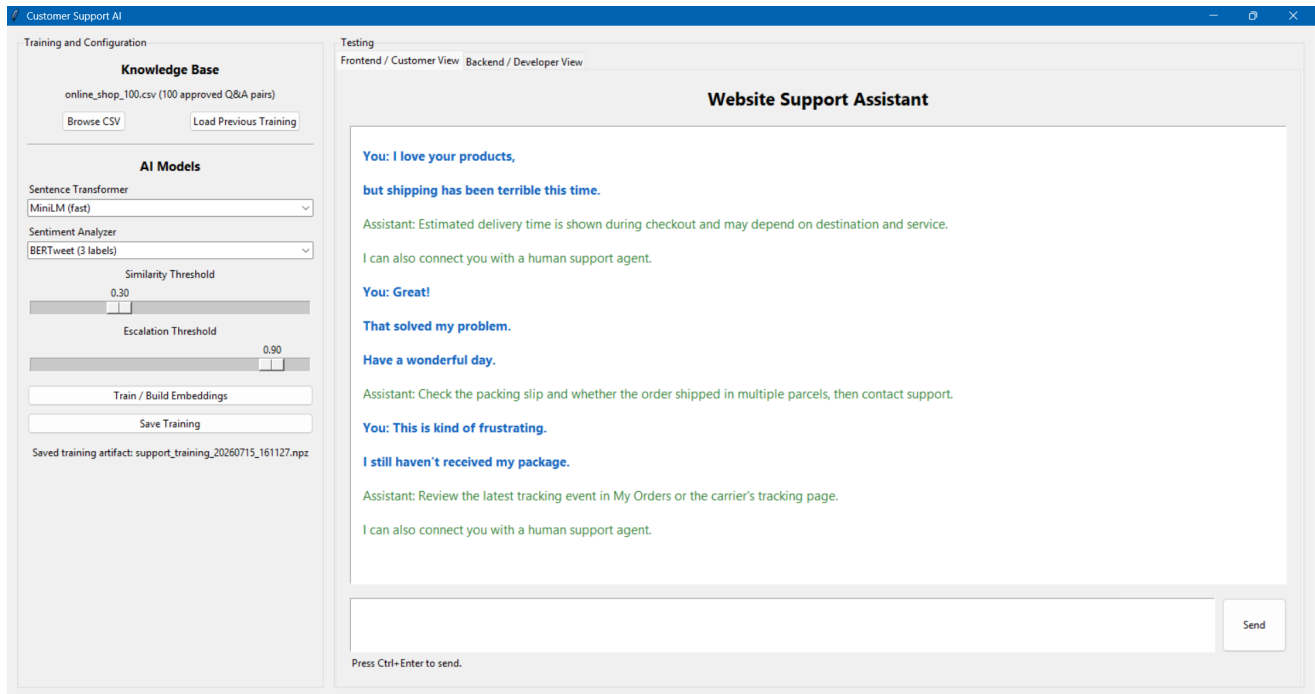
Safety

Because answers always originate from a reviewed knowledge base, the assistant is significantly more predictable than unrestricted text generation. It refuses unsupported questions instead of hallucinating answers.

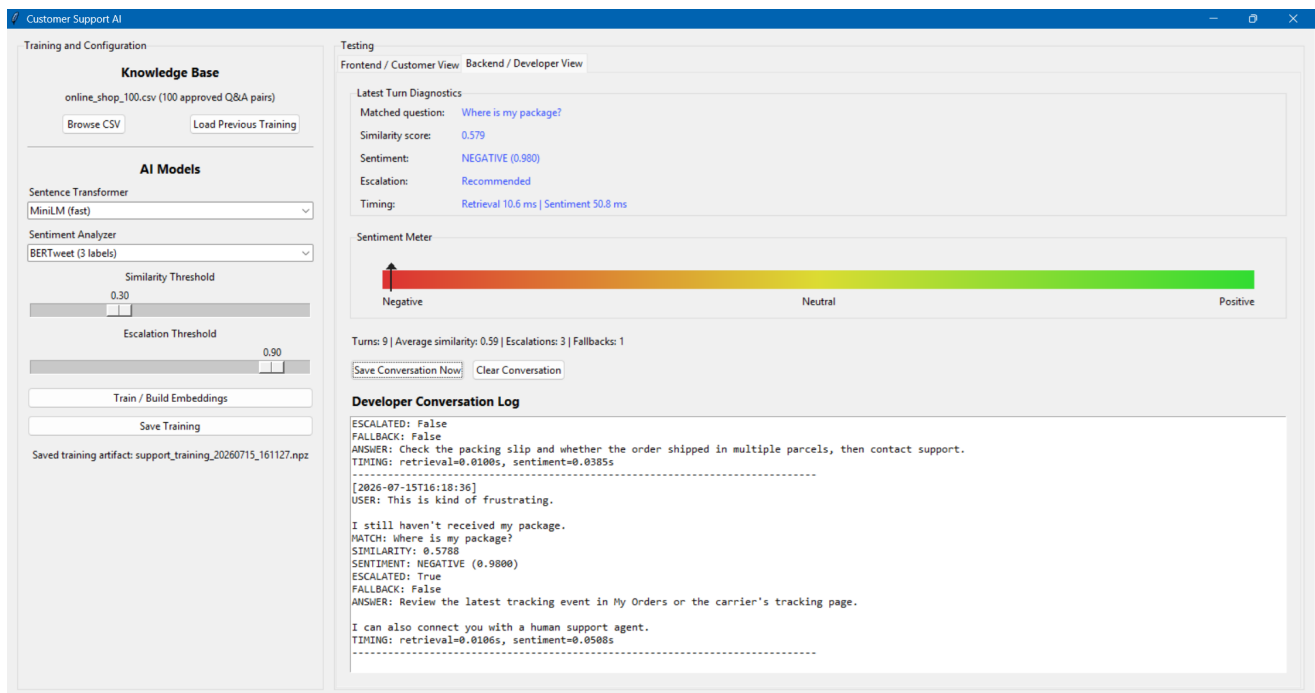
Future Work

Possible extensions include multilingual knowledge bases, live business APIs, conversation memory, web deployment, authentication-aware responses, evaluation datasets, and hybrid retrieval-augmented generation.

Frontend / Customer View



Backend / Developer View



Conclusion

This project demonstrates a complete retrieval-based AI customer support system combining semantic embeddings, cosine similarity search, configurable confidence thresholds, sentiment analysis, escalation logic, persistent training artifacts, and a professional developer/customer interface.